Lecture 17: Managing Missing Data and Linear Regression

Heather Mattie, PhD

October 27, 2026

Recipe of the Day!

Shepherd's Pie



Announcements

- There will be lab this week
- Heather's office hours:
  - Today 2-3pm
  - Thursday 12-1pm
- Midterm due **Friday, November 6th**
  - No late days may be used
- No class November 24th and 26th





Coding question of the day

The **`lung`** dataset in the **`survival`** package has data on 228 patients with advanced lung cancer.

- Create a Table 1 stratified by **`sex`** that includes **`age`**, ECOG performance score (**`ph.ecog`**), Karnofsky score (**`ph.karno`**), weight loss in the past six months (**`wt.loss`**), calories consumed at meals (**`meal.cal`**), and vital status (**`status`**).
- Include an overall column.
- Report **`wt.loss`** and **`meal.cal`** as medians with IQRs.
- Optional extension: show the amount of missing data for each variable as a column in the table.

Be sure to run this code first

```
library(survival)
library(tidyverse)
library(tableone)

lung <- survival::lung %>%
  mutate(sex = fct_recode(as.factor(sex), "Male" = "1", "Female" = "2"),
         status = fct_recode(as.factor(status), "Censored" = "1", "Died" = "2"),
         ph.ecog = as.factor(ph.ecog))
```

Managing missing data

Advanced imputation methods

Heather Mattie, PhD

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Advanced imputation methods

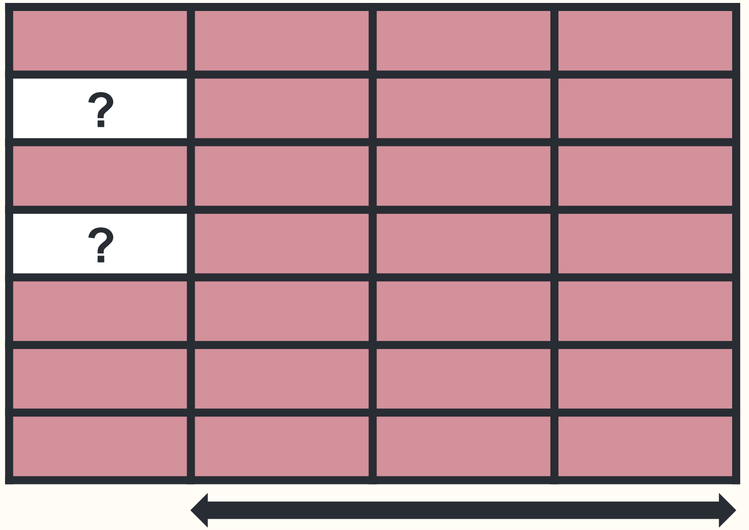
When it can be assumed the type of missingness is MAR, advanced imputation methods are needed

- Methods we will cover include regression imputation and multiple imputation
- These methods take the missingness mechanism into account and reduce some of the bias that imputation can induce

Regression imputation

Regression imputation uses a statistical model to predict what a missing value is likely to be and replaces it with the predicted value

- Other variables in the dataset are used to build the model
- "Regression" means any type of regression model (e.g., linear, logistic, etc.)
- Should be used with caution because variability of imputed data may be underestimated, even if the model is very good

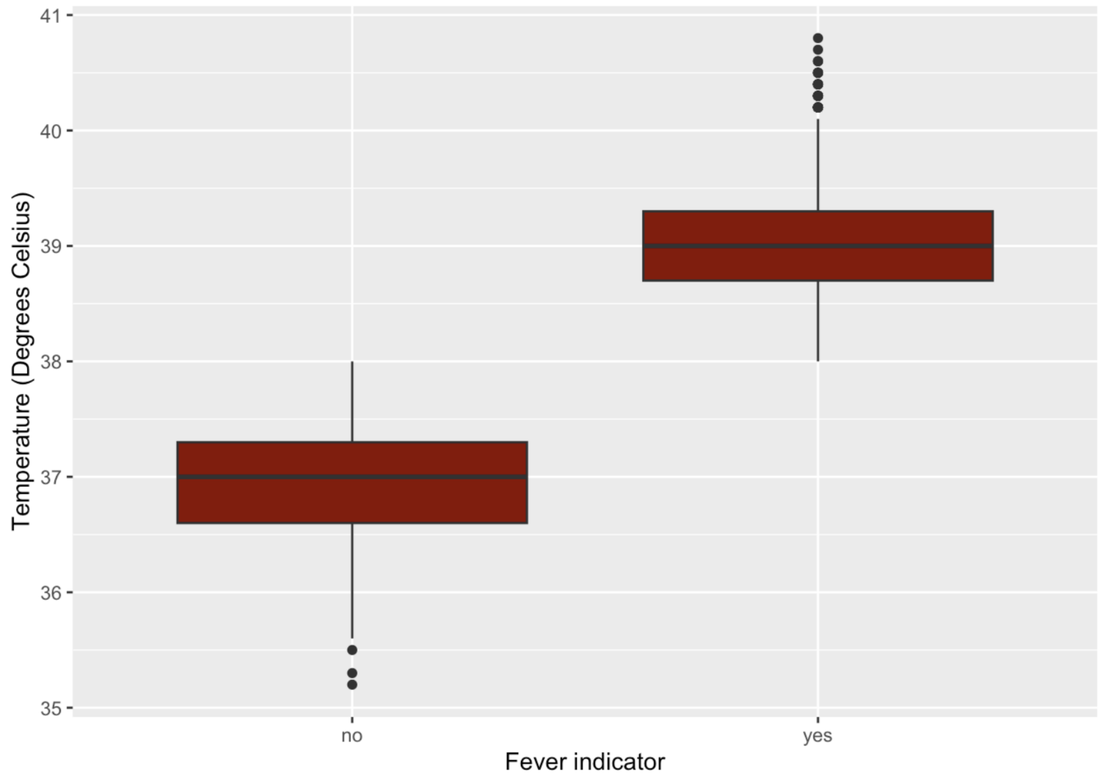


Regression imputation example

Using the Ebola dataset, fever and age will be used in a linear regression model to predict missing values for temperature

- Linear regression is used since temperature is a continuous variable
- Plausible that temperature was not taken if patient presented with chills and aches

```
ebola %>% ggplot(aes(fever, temp)) +
  geom_boxplot(fill = "darkred") +
  xlab("Fever indicator") +
  ylab("Temperature (Degrees Celsius)")
```



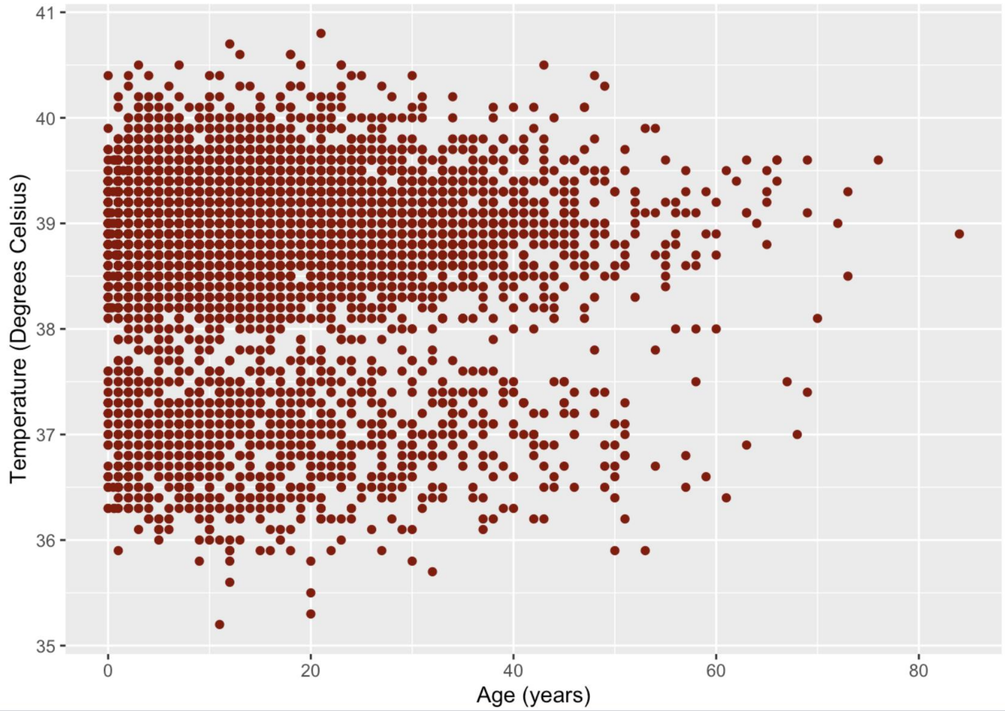
Regression imputation example

Using the Ebola dataset, age and fever will be used in a linear regression model to predict missing values for temperature

- Linear regression is used since temperature is a continuous variable
- Plausible that temperature was not taken if patient presented with chills and aches

```
ebola <- ebola %>%
  filter(!is.na(fever) & !is.na(age_years))

ebola %>% ggplot(aes(age_years, temp)) +
  geom_point(color = "darkred") +
  xlab("Age (years)") +
  ylab("Temperature (Degrees Celsius)")
```



Regression imputation example

Using the Ebola dataset, age and fever will be used in a linear regression model to predict missing values for temperature

- Linear regression is used since temperature is a continuous variable
- Plausible that temperature was not taken if patient presented with chills and aches

```
reg_model <- lm(temp ~ fever + age_years, data = ebola)

newdata <- ebola %>% filter(is.na(temp))

temps_predictions <- predict(reg_model, newdata = newdata)
head(temps_predictions)
```

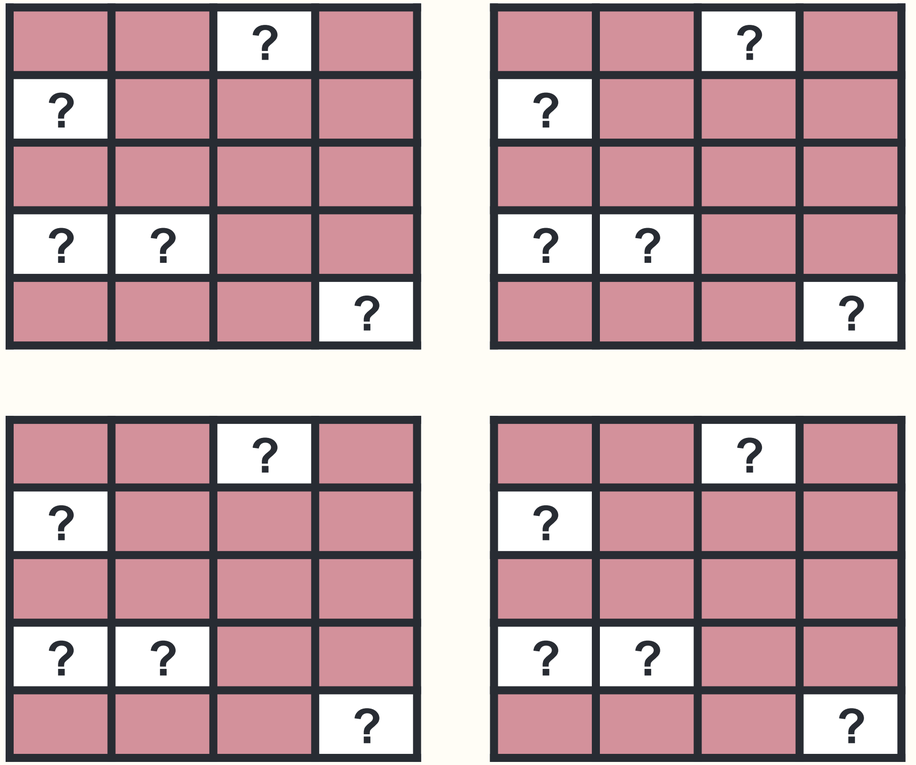
```
##        1        2        3        4        5        6
## 36.97721 36.97175 36.96728 36.97970 36.97622 36.97622
```

Multiple imputation

Why stop with one imputed value per missing value?

Multiple imputation creates multiple datasets with the missing values imputed to plausible data values

- Keep observed values the same in each dataset
- Reduces bias and results in more accurate standard error estimates

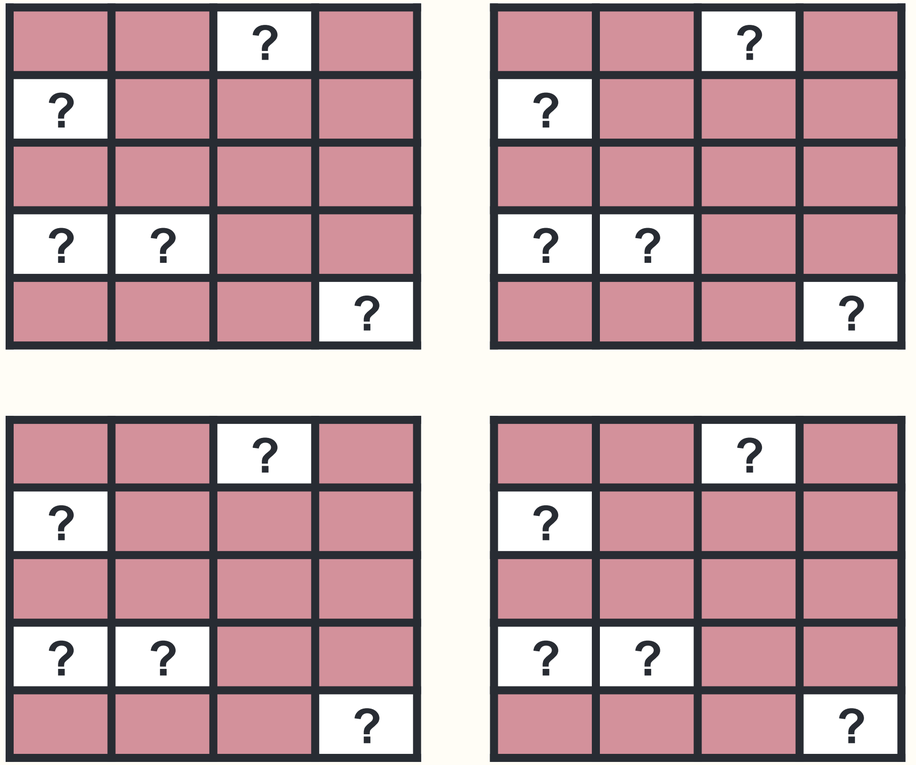


Multiple imputation

Steps:

1. Initialization:

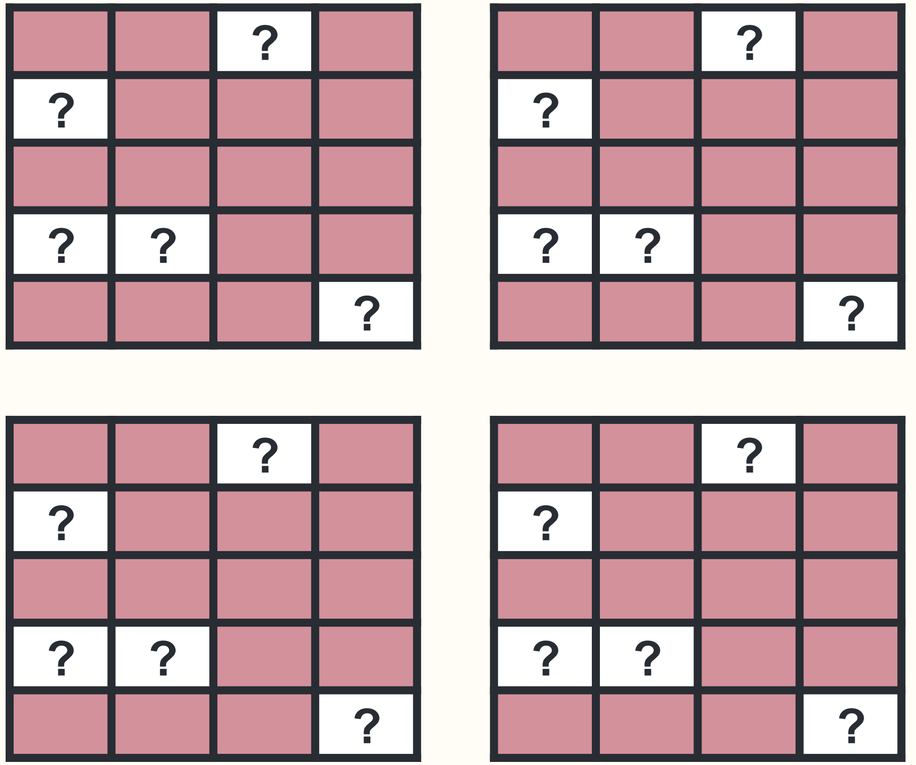
    - Missing values are replaced with plausible values (usually the mean from the observed data)



Multiple imputation

2. Iterative imputation:

- Cycle through each variable with missing values
- Fit a predictive model using the other variables as predictors
- Use model to predict missing values for that variable
- The imputed values are used in subsequent iterations for other variables



Multiple imputation

3. Convergence:

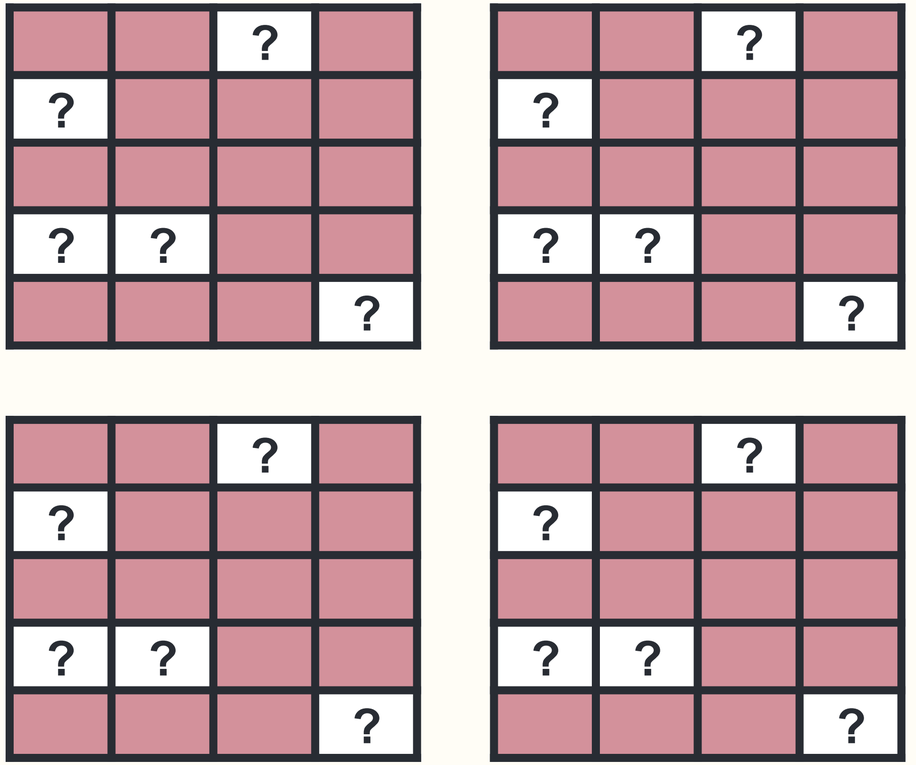
- Iterative process continues until imputed values stabilize

4. Multiple datasets:

- Repeat the process multiple times to achieve multiple datasets

5. Analysis and pooling:

- Analyze each imputed dataset separately, and then pool the results



`mice`

The `mice` function can be used to perform multiple imputation

- Stands for “multiple imputation by chained equations”
- Can impute numerical data, binary data, unordered categorical data, and ordered categorical data

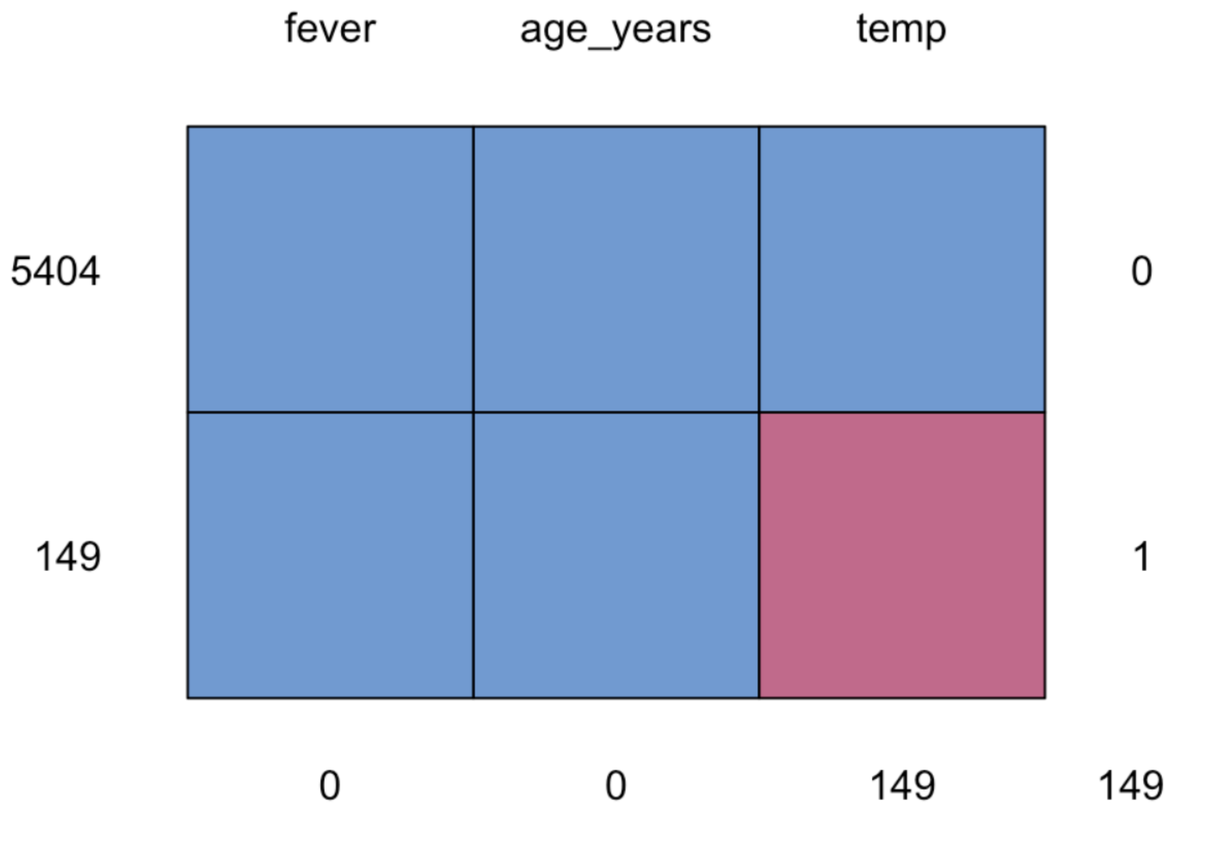


mice

```
library(mice)

model_dataset <- ebola %>%
  select(temp, fever, age_years)

md.pattern(model_dataset)
```



mice

```
multiple_imputation = mice(
  model_dataset,
  seed = 1,
  m = 10,
  print = FALSE)
```

```
model_fit <- with(multiple_imputation, lm(temp ~ age_years + fever))

summary(mice::pool(model_fit))
```

```
##          term     estimate    std.error    statistic        df       p.value
## 1 (Intercept) 3.702800e+01 0.0258516265 1432.3277729  31.99099  1.797488e-78
## 2   age_years 2.344859e-04 0.0006359435    0.3687213 110.48345  7.130411e-01
## 3    feveryes 1.978807e+00 0.0191893059  103.1203259 227.16366 9.200270e-193
```

mice

```
model_fit <- with(multiple_imputation, lm(temp ~ age_years + fever))

summary(mice::pool(model_fit))
```

```
##          term     estimate    std.error    statistic        df       p.value
## 1 (Intercept) 3.702800e+01 0.0258516265 1432.3277729  31.99099  1.797488e-78
## 2   age_years 2.344859e-04 0.0006359435    0.3687213 110.48345  7.130411e-01
## 3    feveryes 1.978807e+00 0.0191893059  103.1203259 227.16366 9.200270e-193
```

```
complete_data <- model_dataset[complete.cases(model_dataset),]

model_fit <- with(complete_data, lm(temp ~ age_years + fever))

tidy(summary(model_fit))
```

```
## # A tibble: 3 × 5
##   term         estimate std.error statistic p.value
##   <chr>           <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept) 37.0       0.0168    2201.       0
## 2 age_years    0.000497  0.000506     0.982    0.326
## 3 feveryes     2.05      0.0163     126.       0
```

Summary

- When data is MAR, advanced imputation methods should be used
- Regression imputation uses a statistical model to impute missing values
- Multiple imputation creates multiple datasets with imputed values and uses them to calculate pooled estimates for analysis
- The `mice` function can be used to perform multiple imputation
- Both methods reduce bias and result in more accurate estimates

Managing missing data

Survey skip patterns

Heather Mattie, PhD

Surveys

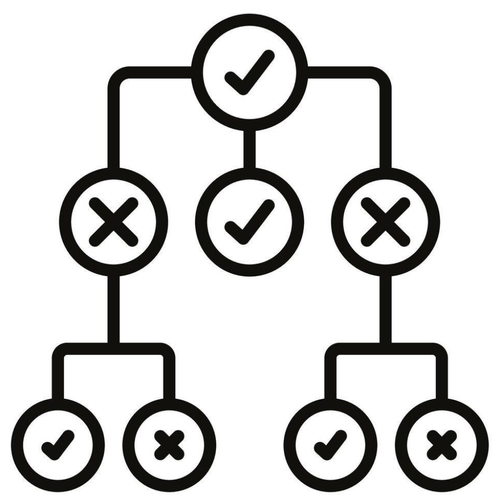
Public health surveys are vital tools for understanding population health and informing public health policy



- Collect data on health status, behaviors, risk factors, etc.
- Allow researchers and policymakers to identify health disparities, track trends, and evaluate interventions
- Can be conducted at the national, state, or local level
- Can involve a combination of questionnaires, interviews, physical examinations, and lab tests

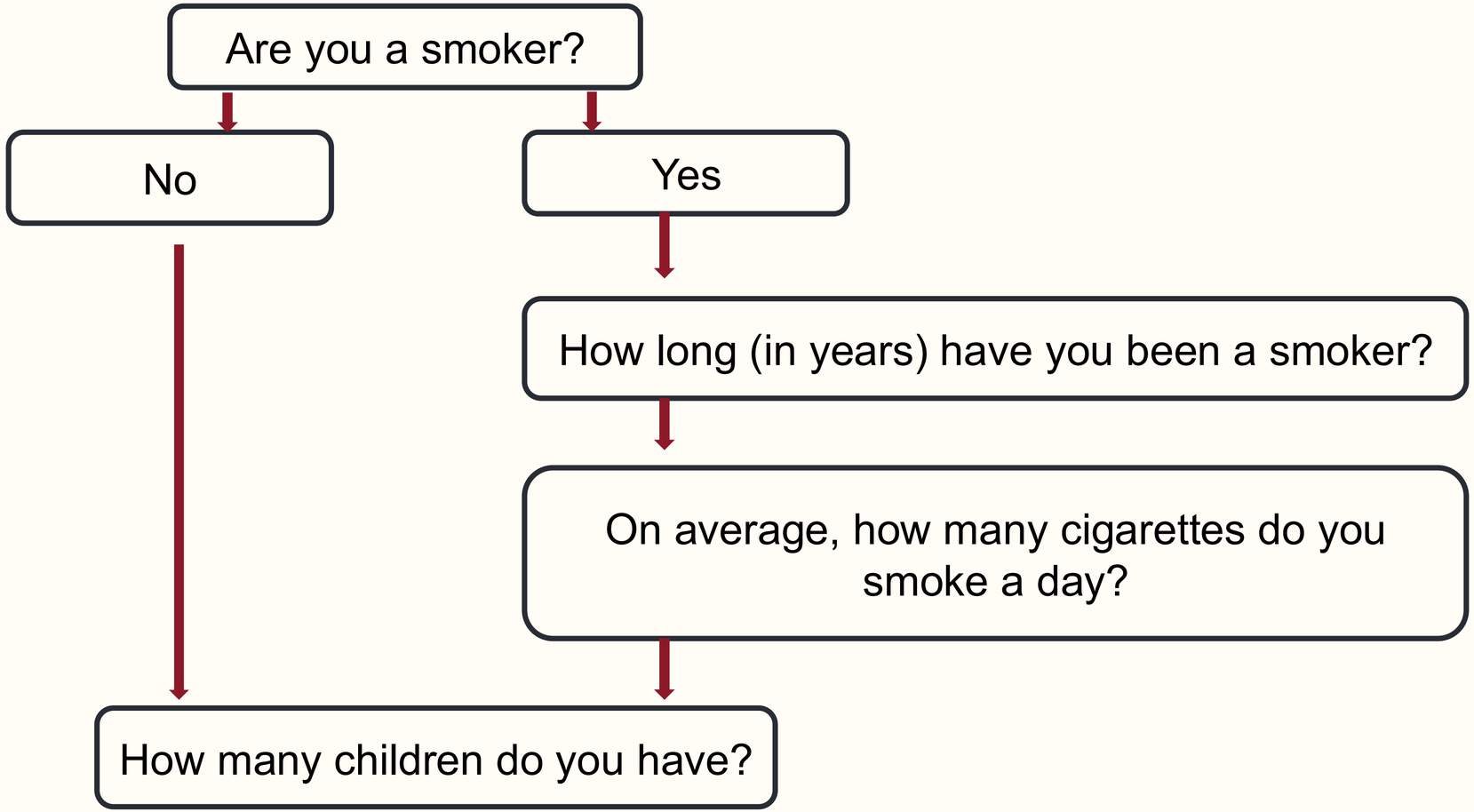
Skip pattern

A skip pattern in a survey is a feature that directs respondents to different questions or sections of a survey based on their answers



- Also called “skip logic” or “conditional branching”
- Ensures respondents only answer questions relevant to their specific situation
- Improves survey efficiency and engagement through streamlining
- Induces missingness by design

Skip pattern example



Skip pattern example

```
data
```

```
##   smoke smoke_years smoke_amt children
## 1   Yes           24        20        0
## 2   Yes            7        15        1
## 3    No           NA        NA        4
## 4    No           NA        NA        2
```

Skip pattern example

```
data
```

```
##   smoke smoke_years smoke_amt children
## 1   Yes           24        20        0
## 2   Yes            7        15        1
## 3    No           NA        NA        4
## 4    No           NA        NA        2
```

```
data <- data %>%
  mutate(smoke_years = ifelse(is.na(smoke_years), 0, smoke_years),
         smoke_amt = ifelse(is.na(smoke_amt), 0, smoke_amt))

data
```

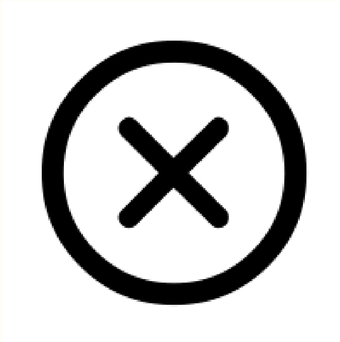
```
##   smoke smoke_years smoke_amt children
## 1   Yes           24        20        0
## 2   Yes            7        15        1
## 3    No            0         0        4
## 4    No            0         0        2
```

Response bias

Response bias occurs when a respondent answers questions incorrectly or in a way that deviates from their true opinions or behaviors



- Many causes of response bias, we will focus on lack of understanding and accidental incorrect response
- Rather than removing these individuals from the dataset, can sometimes replace values that are known to be incorrect



Response bias example

Survey sent out to students at a university in the United States

- Included a question that asked if the respondent had received a federal grant
- This particular grant is only awarded to domestic students
- All international students *should* respond "No" to the federal grant question



Response bias example

```
data
```

```
##   international        grant
## 1           Yes I don't know
## 2           Yes          Yes
## 3           Yes           No
## 4            No          Yes
## 5            No           No
```

Response bias example

```
data
```

```
##   international        grant
## 1           Yes I don't know
## 2           Yes          Yes
## 3           Yes           No
## 4            No          Yes
## 5            No           No
```

```
data <- data %>%
  mutate(grant = ifelse(international == "Yes" &
                          grant != "No", "No", grant))
data
```

```
##   international grant
## 1           Yes    No
## 2           Yes    No
## 3           Yes    No
## 4            No   Yes
## 5            No    No
```

Summary

- Surveys are a common source of data in public health
- Survey skip patterns improve survey efficiency, but induce missingness by design
- Most skip pattern missing values can be imputed in a straightforward and logical way
- Response bias should be checked for before removing individuals from a dataset

Linear Regression

Heather Mattie, PhD

HARVARD T.H. CHAN
SCHOOL OF PUBLIC HEALTH

Linear Regression Roadmap

1. Correlation
2. Simple linear regression
3. Evaluation of model assumptions
4. Multivariate linear regression
5. Collinearity
6. Model selection

Linear Regression

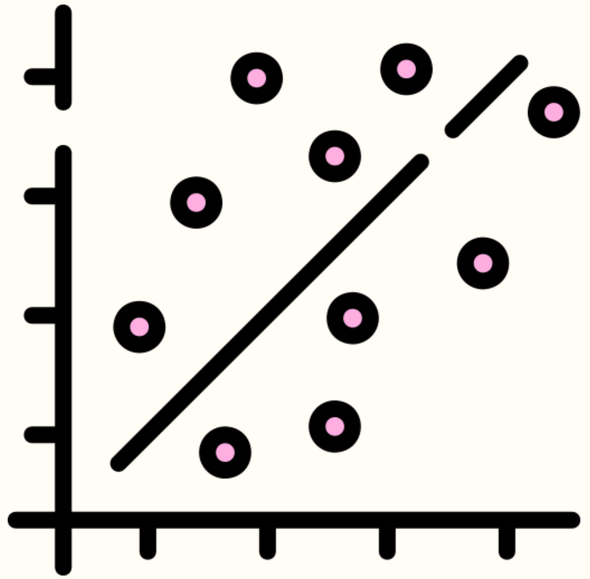
Correlation

Heather Mattie, PhD

Correlation

Correlation quantifies the **linear** relationship between two continuous variables

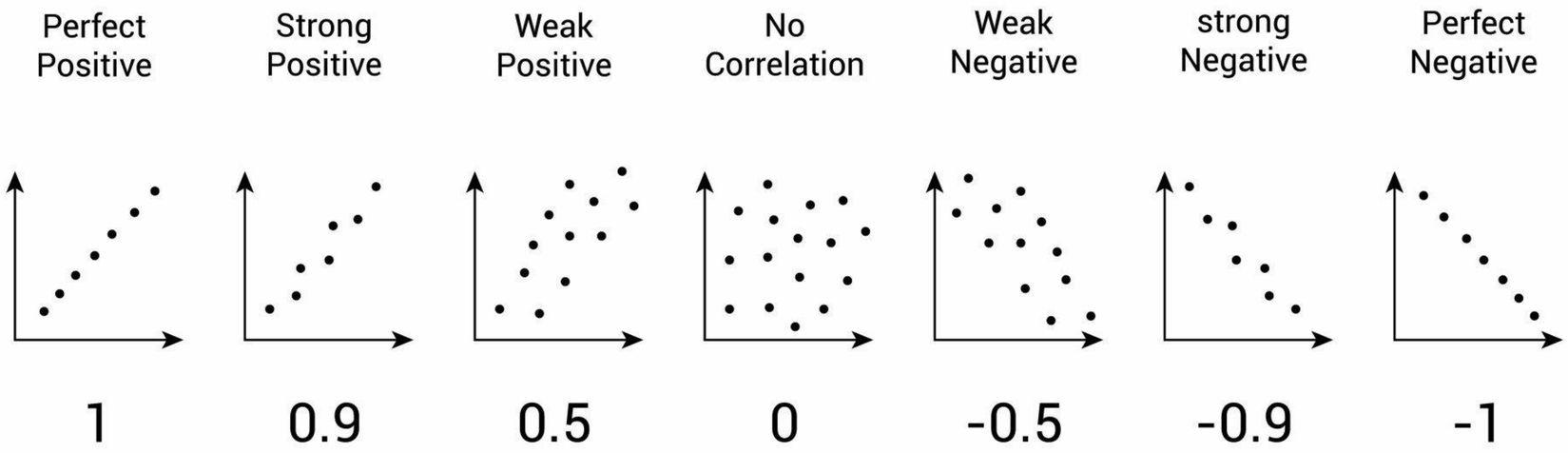
- Can measure the magnitude of the association between the two variables
- Scatterplots help visualize the relationship
- Two correlation estimates: Pearson correlation coefficient and Spearman rank correlation coefficient



Correlation

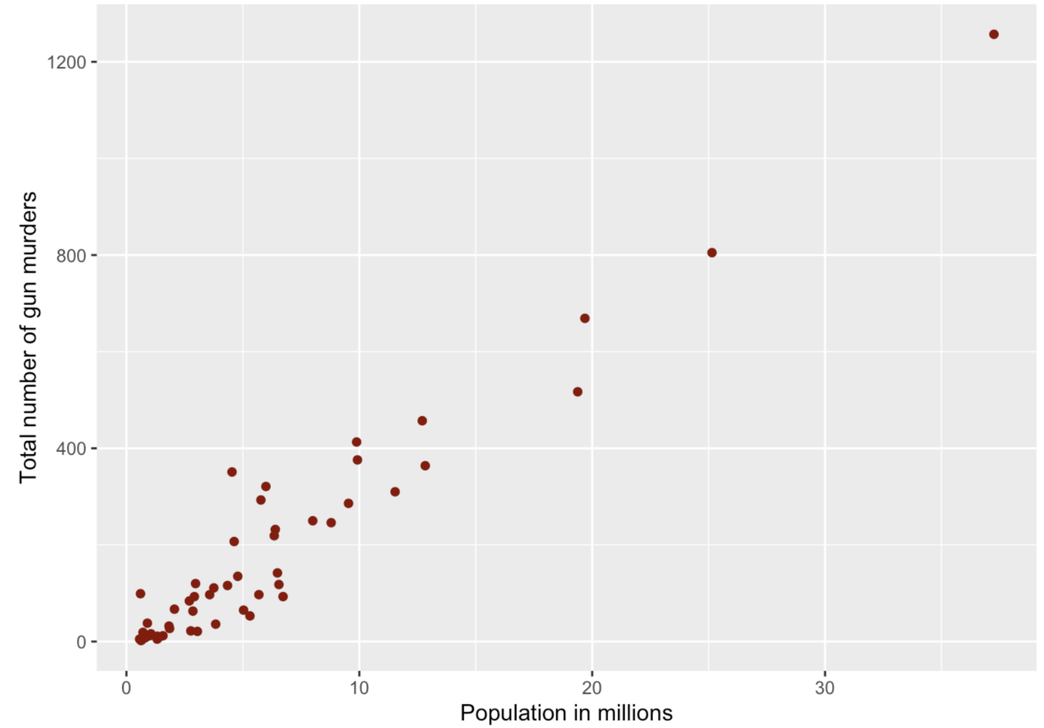
Correlation ranges from -1 to 1 and has no units of measurement

- Values closer to 1 or -1 represent a stronger linear relationship
- A value of 0 means no **linear** relationship



Example: gun murders

```
murders %>%
  ggplot(aes(population/10^6, total)) +
  geom_point(color = "darkred") +
  xlab("Population in millions") +
  ylab("Total number of gun murders")
```



Pearson correlation coefficient

The Pearson correlation coefficient (r) is a point estimate of ρ (the population correlation)

Assumption:

- Both variables are approximately normally distributed

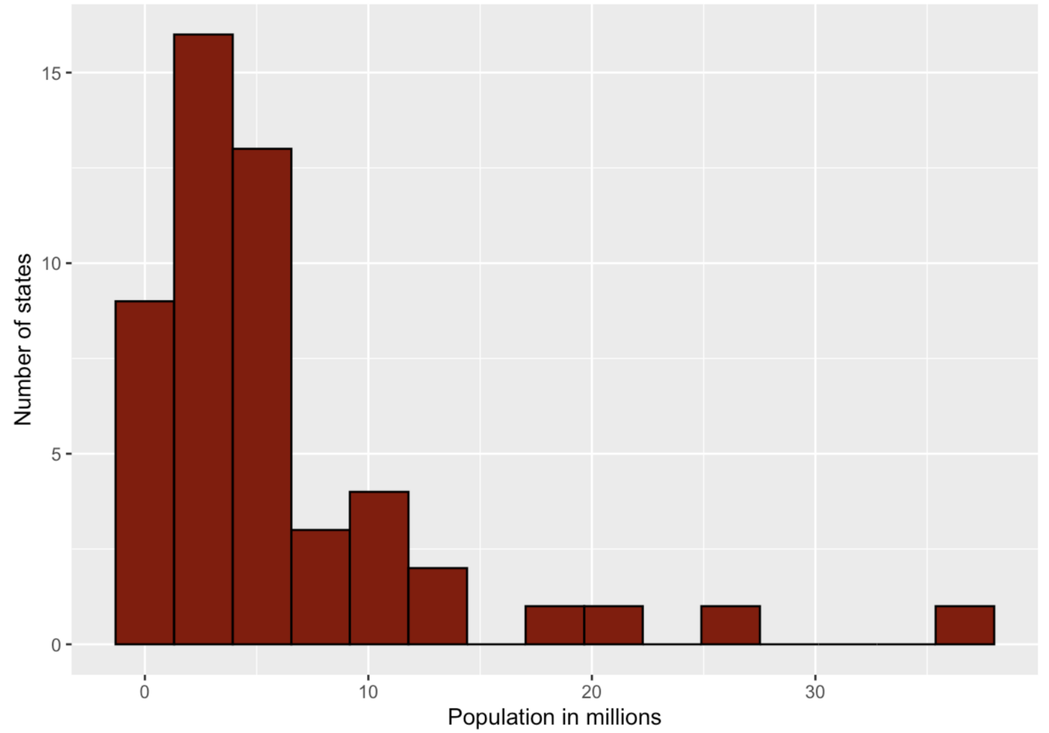
Pearson correlation coefficient

The Pearson correlation coefficient (r) is a point estimate of ρ (the population correlation)

Assumption:

- Both variables are approximately normally distributed

```
murders %>%
  ggplot(aes(population/10^6)) +
  geom_histogram(color = "black", fill = "darkred", bins = 15) +
  xlab("Population in millions") +
  ylab("Number of states")
```



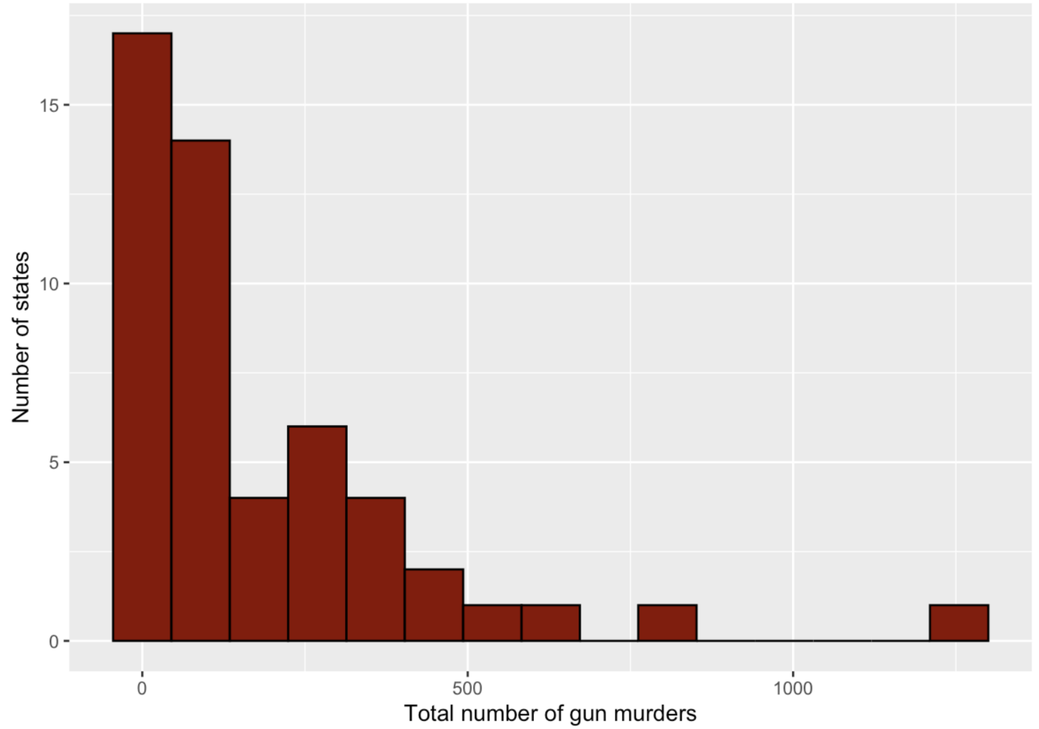
Pearson correlation coefficient

The Pearson correlation coefficient (r) is a point estimate of ρ (the population correlation)

Assumption:

- Both variables are approximately normally distributed

```
murders %>%
  ggplot(aes(total)) +
  geom_histogram(color = "black", fill = "darkred", bins = 15) +
  xlab("Total number of gun murders") +
  ylab("Number of states")
```



Pearson correlation coefficient

The Pearson correlation coefficient (r) is a point estimate of ρ (the population correlation)

Assumption:

- Both variables are approximately normally distributed

```
cor(murders$population, murders$total)
```

```
## [1] 0.9635956
```

Pearson correlation coefficient

Null hypothesis: $\rho = 0$

Alternative hypothesis: $\rho \neq 0$

Conclusion: the population correlation is significantly different than 0.

```
cor.test(murders$population, murders$total)
```

```
##
##  Pearson's product-moment correlation
##
## data:  murders$population and murders$total
## t = 25.228, df = 49, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.9367732 0.9791617
## sample estimates:
##       cor
## 0.9635956
```

Spearman rank correlation coefficient

The Spearman correlation coefficient (r_S) is a point estimate of ρ (the population correlation)

- Does not assume both variables are approximately normally distributed
- Nonparametric; based on ranks
- Less sensitive to outliers and skewed distributions

```
cor(murders$population, murders$total, method = "spearman")
```

```
## [1] 0.8906688
```

Spearman rank correlation coefficient

Null hypothesis: $\rho = 0$

Alternative hypothesis: $\rho \neq 0$

Conclusion: the population correlation is significantly different than 0.

```
cor.test(murders$population, murders$total, method = "spearman")
```

```
##
##  Spearman's rank correlation rho
##
## data:  murders$population and murders$total
## S = 2416.2, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.8906688
```

Example: Prenatal care

In 1989, Seattle and King County in Washington state began a program called First Steps that provides free prenatal care to low-income women.

- Includes both medical attention and advice about lifestyle decisions that may affect birthweight
- The goal of First Steps is to improve birth outcomes, including lower prevalence of low birthweight babies



Example: Prenatal care

`firstep`: indicates participation in First Steps

`gestation`: gestational age at birth (in weeks)

`age`: the mother's age at birth

`wpre`: mother's pre-pregnancy weight in pounds

`wgain`: amount of maternal weight-gain during pregnancy

`education`: years of mother's education

`bwt`: birthweight (grams)



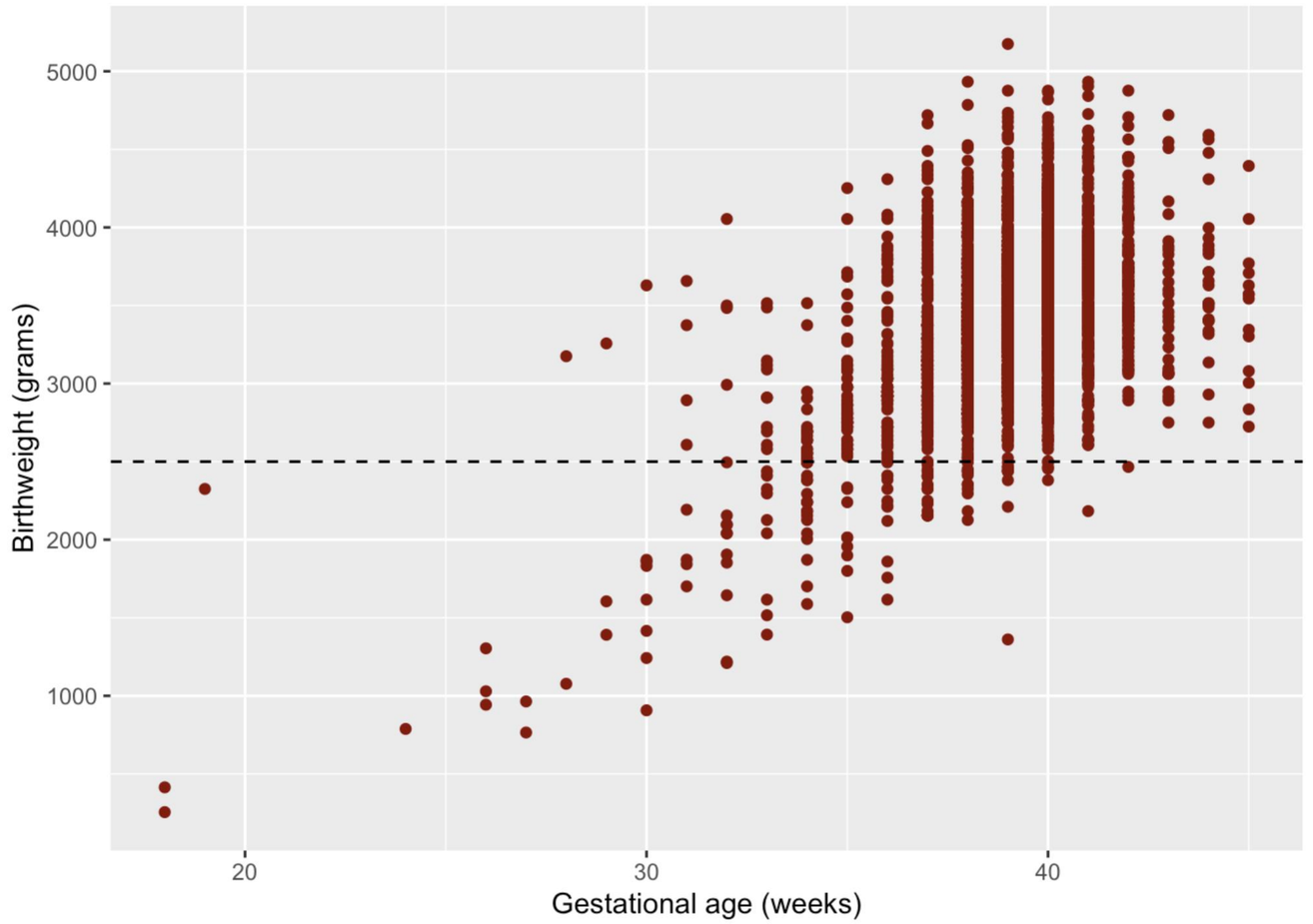
Example: Prenatal care

```
data <- read.table("KingCounty2001_data.txt", header = TRUE)
head(data)
```

```
##   gender plural age     race parity married  bwt smokeN drinkN firstep welfare
## 1      F      1  31 hispanic      4       1 3118      0      0       1       0
## 2      M      1  23    white      0       1 3466      0      0       1       0
## 3      F      1  24    black      1       0 3147      0      0       1       0
## 4      M      1  26 hispanic      2       1 3969      0      0       1       0
## 5      M      1  34    white      0       0 3005      0      0       1       0
## 6      M      1  15    black      0       0 2920      0      0       1       0
##   smoker drinker wpre wgain education gestation
## 1      N       N  122    22         5        40
## 2      N       N  160    50        12        39
## 3      N       N  150    38        13        40
## 4      N       N  175    15        12        39
## 5      N       N  123    17        17        40
## 6      N       N  180    12         9        36
```

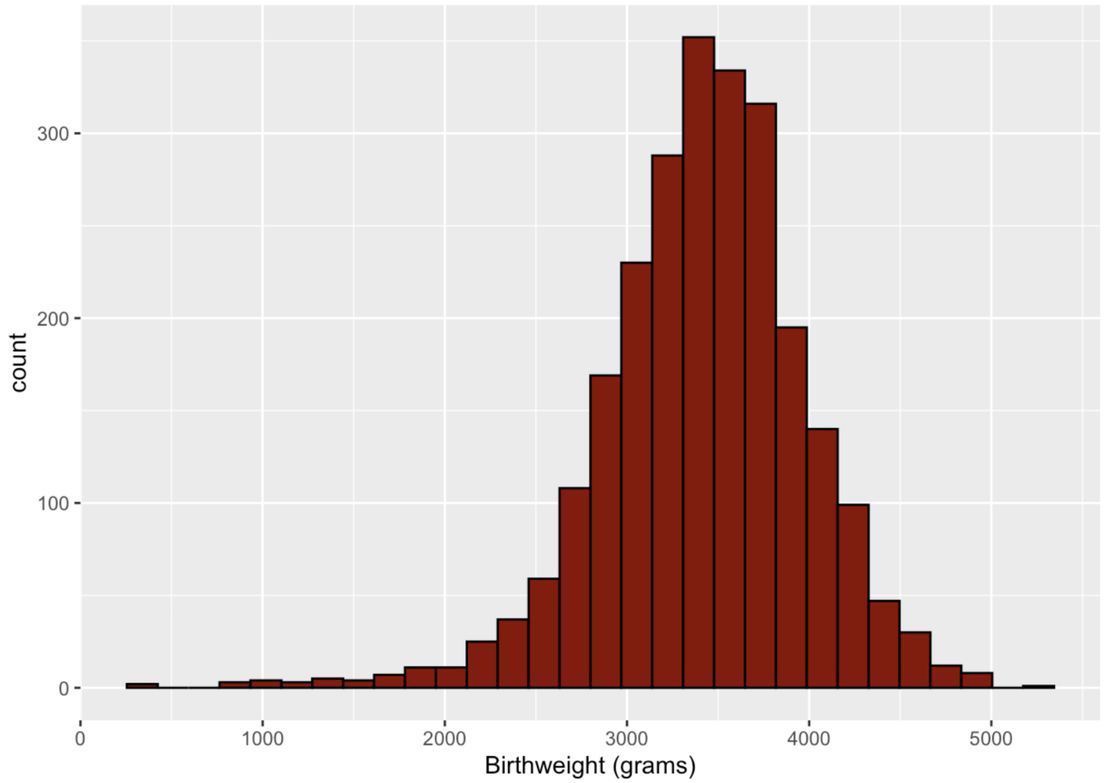
Example: Prenatal care

```
data %>% ggplot(aes(gestation, bwt)) +
  geom_point(color = "darkred") +
  xlab("Gestational age (weeks)") +
  ylab("Birthweight (grams)") +
  geom_hline(yintercept = 2500, lty = 2)
```

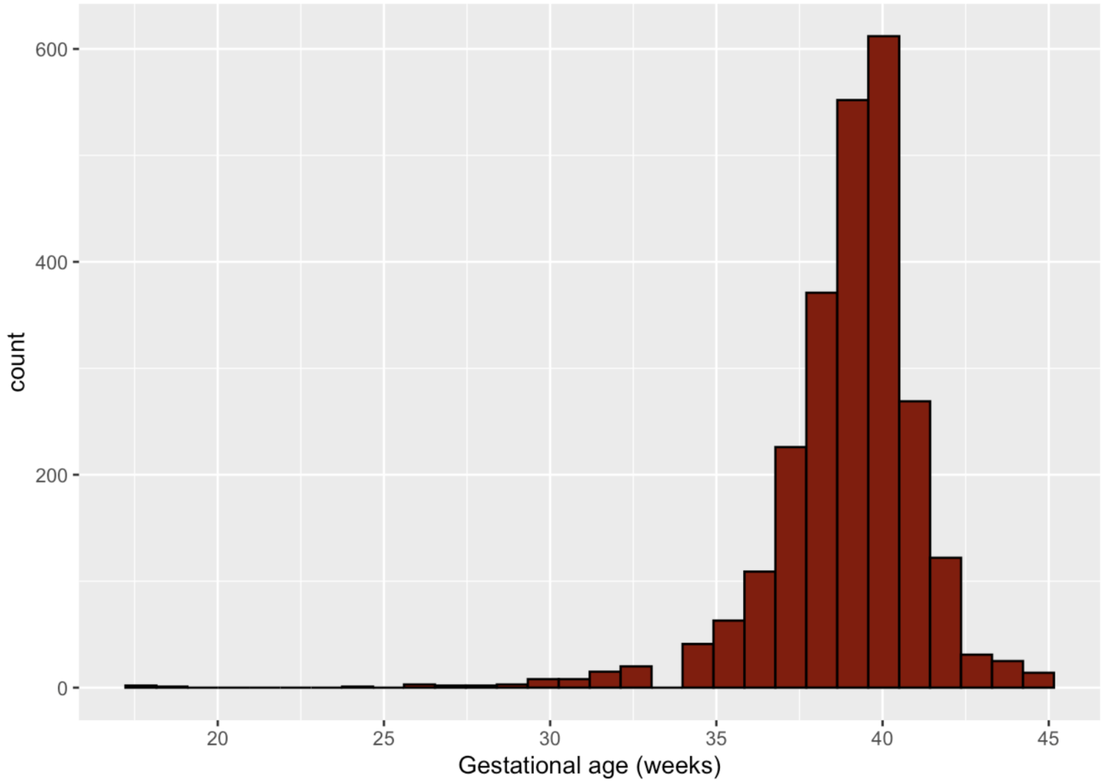


Example: Prenatal care

```
data %>% ggplot(aes(bwt)) +
  geom_histogram(color = "black", fill = "darkred") +
  xlab("Birthweight (grams)")
```




```
data %>% ggplot(aes(gestation)) +
  geom_histogram(color = "black", fill = "darkred") +
  xlab("Gestational age (weeks)")
```



Example: Prenatal care

```
cor(data$gestation, data$bwt)
```

```
## [1] 0.527407
```

```
cor.test(data$gestation, data$bwt)
```

```
##
##  Pearson's product-moment correlation
##
## data:  data$gestation and data$bwt
## t = 31.026, df = 2498, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.4985114 0.5551319
## sample estimates:
##      cor
## 0.527407
```

```
cor(data$gestation, data$bwt, method = "spearman")
```

```
## [1] 0.4093581
```

```
cor.test(data$gestation, data$bwt, method = "spearman")
```

```
##
##  Spearman's rank correlation rho
##
## data:  data$gestation and data$bwt
## S = 1538129750, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.4093581
```

Summary

- Correlation quantifies the linear relationship between two continuous variables
- The Pearson correlation coefficient r is sensitive to outliers and requires the two continuous variables of interest to be approximately normally distributed
- If one or both variables are not approximately normally distributed the Spearman correlation r_s should be used instead